

Referee report on *A two-stage stochastic programming approach incorporating spatially-explicit fire scenarios for optimal firebreak placement*

This paper proposes a two-stage stochastic model for the optimal location of firebreak in a forest under wildfires. A rectangular grid represents the forest, and an initial wildfire starts in a random cell of this grid. Using state-of-the-art fire simulators, it generates a set of random scenarios representing the propagation of the wildfire through a directed graph rooted at the starting cell. The proposed model selects the optimal set of cells to use as firebreaks, avoiding fire propagation through the directed graph for each scenario. In addition, the authors suggest optimizing a combination of the expected cost of forest damage and a risk measure (CVaR) to reinforce the impact of the worst scenarios. The model is solved using its straightforward formulation using Gurobi. Computational experiments show that the model can solve an instance of 20x20 cells with 60 scenarios up to optimality. However, increasing the number of scenarios and the number of allowed firebreaks became harder to solve. Further analysis is done for the solutions obtained for this case, evaluating the robustness of the solutions using out-of-sample simulations assuming a static and dynamic fire spread behavior.

The paper's subject is very interesting, and computational experiments show considerable potential for saving when optimally locating firebreaks. The proposed model is simple and straightforward but adequate for the problem. The main weakness of the paper is its methodology to solve the problem. Solving the formulation on an MIP solver can only deal with problems of limited size. As the authors show in their analysis, it is imperative to consider a large number of scenarios to represent the stochasticity of the problem. On the other hand, the structure of the problem allows us to solve it much more efficiently using, for example, decomposition algorithms. The analysis of the obtained solutions is interesting, showing the impact of the solution and opening many exciting questions to analyze.

Due to the high standard of the EJOR journal, this paper cannot be accepted in its present form. As mentioned, the methodology is simplistic, becoming the major obstacle to being applicable in practice. Nevertheless, there is significant room for improvement because solving the problem can be enhanced using standard decomposition techniques for stochastic problems. I'll be pleased to review a revision of this paper adding a stronger methodology able to solve larger instances.

Major comments

Improvements to the methodology: It is hard to understand why not to use decomposition techniques since this problem is very appropriate for these techniques. A deeper analysis of the stochastic subproblem shows these required properties. In fact, these sub-problems can be solved without needing an MIP solver, so it can be used to evaluate the solutions (Section 4.2) for an even more significant number of scenarios.

- **Subproblem formulation:** Many simple changes can improve the second-stage problem. First, the integrality of the x variables is not required, and it can be replaced by simply $x \geq 0$. In fact, because of the positive coefficients on the objective function, the largest value for x variables is given by the cell forced to be one, so no further upper bounds are needed.
- **Subproblem solution for a given y assignment:** Solving the subproblem for a given y can be done directly. Given the directed graph of fire spread, the variables with value $x = 1$ are simply the nodes reachable from the root node (eliminating the nodes where a fireplace has been installed). This can be done using a BFS traversal of this graph in linear time. After this, the value of each scenario can be computed, and the CVaR will be the average of the 10% worst cases. If the explicit solution is required, this is equivalent to setting the variable θ to the value of the 10%-scenario and setting ν^w to the difference between the value of the scenario and θ . This methodology should be a way of evaluating a given solution (particularly in Section 4) because it is more efficient and less expensive than using an MIP model for solving this trivial problem.

- **Dual formulation of the Subproblem and Bender cuts:** Since the Subproblem is trivially solvable, and the integrality of the second-stage variable is not required, it is possible to derive a Bender formulation for this problem. It requires modifying the second-stage problem to have consistency of the subproblem for each scenario. For example, you can replace constraints (8c) and (8d) by:

$$\begin{aligned} w_v^\omega &\geq \epsilon_v^\omega & v \in \mathcal{V}, \omega \in \Omega \\ w_v^\omega &\leq w_u^\omega + y_u + \xi_{uv}^\omega & uv \in \mathcal{E}, \omega \in \Omega \end{aligned}$$

where $\epsilon_v^\omega = 1$ iff the fire starts at node v on scenario ω , and $\xi_{uv}^\omega = 0$ iff arc $(u, v) \in \mathcal{E}^\omega$. In this way, we obtain an equivalent formulation for the second-stage subproblem that does not change for each scenario (except for the random coefficients ϵ, ξ).

The dual of this problem is easy to write, and the dual optimal solution can also be obtained analytically from the primal solution of the subproblem discussed previously. Note that, in this case, it is even simpler because γ is a (fixed) first-stage variable. Hence, a Bender cut can be easily obtained for each scenario (probably in linear time). This decomposition method allows us to avoid all the second-stage variables, reducing the problem considerably. On the other hand, several recent results exist for Benders cuts using adaptive aggregation of scenarios. These methods work particularly well with CVaR objectives, making it possible to solve problems with an even more significant number of scenarios.

Simulation of the fire spreads. As described, the simulation of fire spread from the software are directed acyclic graph (DAG). However, figures show a more restricted structure than a DAG, a rooted tree, with a single path from the starting fire to each cell. Is this correct?

It is interesting to model the fire spread as a directed graph, but the assumption of being a tree is too strong and simplistic. It is hard to believe that installing a firebreak in a node automatically protects all the nodes in the sub-tree below the firebreak node. The analysis of the solutions under static and dynamic fire spread apparently confirms this issue. If this is the case, is it possible to improve the simulations to cover multiple paths for each cell? (for example, merging several simulations into a single one). This could improve the gap in the robustness between the static and dynamic scenario.

A more reasonable assumption is that the graph is a directed acyclic graph (so there are several paths to spread the fire from the source to a given cell). Also, the proposed MIP model covers the case of a general DAG (moreover, the same formulation also applies to non-acyclic directed graph). If my interpretation is incorrect and the simulations are effectively DAGs (and not just rooted trees), please correct Figure 1 and 2 to represent the true DAG behind the fire spread.

Minor comments

- Page 4, line -5: $L(y; e; \tau)$ is not a *random variable that represents the burned cells*. It is a the **value** of the loss due to the burned cells (which is a random variable).
- Page 4 and 6: Please use the same notation for the randomness between these two sections. In Page 4, the random parameter is τ and the set of scenarios is T (and in equation 3 it appears as $\mathcal{S}$). Later (page 6) the random variable is ω and the set of scenarios is Ω .
- Page 5, equation (3). The discussion about this formula assumes that scenarios are i.i.d so equiprobable. But the next section is more general, assuming arbitrary probabilities for each scenario. I suggest to choose one of them, and be consistent between the two sections.
- Page 5, Figure 1 and 2. In my understanding, each of the tree plots has a different scenario (the directed graph in red is different in the three cases). In this case, it does not make sense to compare the resulting value of different solutions (point iii), because even the same solution y will have different objective values in each case. Either make this comparison but using the same scenario, or avoid these meaningless comparison of the objective values.

- Page 6, line -11: "*subject to α* ". Rewrite this in a formal way.
- Page 7, Section 3.1. I suggest to move this subsection to the next section.
- Page 9, Subsection 4.2 "*a test set comprising $\mathcal{S}$ simulations*". $\mathcal{S}$ is the set (not the number of simulations). Also I suggest to call this set as "out-of-sample" scenarios.
- Page 9, middle fo the page: "*These graphs are incorporated into the mathematical formulation by the.*". This line is incomplete. Also remark that you do not need to solve a MIP model for evaluating a solution.
- Page 9, Figure 3. The figure is not clear enough. I suggest to remove it, or add a text explaining the figure. For example, is not clear which box is a data, a process, or a result.
- Page 10. Why to introduce a random solution for comparison? It is obvious that this solution will perform very bad, so it does not contribute to the discussion. More importantly, a single random first-stage decision depends on the random realization (it could be the optimal solution, for example). I suggest to remove this. If not, please add not only one random sample first-stage decision, but a sufficiently large number of them, to validate these values.
- Page 12, caption of Figure 5c: Typo "t"
- Page 12, Figure 5. The fact that the presented solutions are not the optimal solution, but the best solution found before the time limit, can invalidate the conclusions obtained. I suggest to restrict to solutions with proved optimality (may be increasing the time limit)